OSHA 30-HOUR CONSTRUCTION

Comprehensive Ultimate Safety Examination Manual • 27 Core Chapters

Welcome to Your Ultimate Exam Preparation Guide: This comprehensive technical manual condenses the full 30-Hour OSHA Construction Safety syllabus into an easy-to-understand, highly structured study framework. Designed specifically for last-minute cramming and deep content review before your professional safety exam, this document highlights critical threshold triggers, numerical standards, and high-yield testing areas across all 27 core instructional modules.

Table of Study Chapters

Chapter 1: Introduction to OSHA and the OSH Act

General Provision & History

The Occupational Safety and Health Administration (OSHA) was created in 1970 under the Occupational Safety and Health Act signed by President Richard Nixon. Its primary purpose is to ensure safe and healthful working conditions for workers by setting and enforcing standards and by providing training, outreach, education, and assistance. A fundamental pillar of the OSH Act is the General Duty Clause, Section 5(a)(1), which requires employers to provide their employees with a place of employment that is free from recognized hazards that are causing or are likely to cause death or serious physical harm.

MANDATORY STANDARDS & TECHNICAL CORE

- Workers have the right to a safe workplace, to raise health and safety concerns without fear of retaliation, and to receive information and training about hazards.
- Section 11(c) of the OSH Act prohibits any person from discharging or in any manner discriminating against any employee because they exercised their rights under the act.
- Employers must provide PPE, maintain records of work-related injuries, and display the official OSHA poster in a prominent location.
- Workers have the right to request an OSHA inspection and participate in it, as well as access records of workplace injuries and illnesses.

> **★ HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
> *Remember Section 5(a)(1) - The General Duty Clause. It is invoked when no specific standard exists for a clear, recognized workplace hazard. Also note that whistleblower protection falls under Section 11(c).*

Chapter 2: OSHA Inspection Procedures and Citations

29 CFR 1903

OSHA enforcement involves highly structured workplace inspections conducted by Compliance Safety and Health Officers (CSHOs). Inspections are conducted without advance notice, with rare exceptions. OSHA prioritizes its inspections based on severity: 1st priority goes to Imminent Danger situations; 2nd priority goes to Fatalities and Catastrophes (events resulting in hospitalizations); 3rd priority is Employee Complaints and Referrals; and 4th priority is Programmed Inspections targeting high-hazard industries.

MANDATORY STANDARDS & TECHNICAL CORE

- The inspection process consists of four distinct phases: Presentation of Credentials by the CSHO, the Opening Conference, the Walkthrough Inspection, and the Closing Conference.
- During the walkthrough, the compliance officer may interview employees privately and inspect safety programs and logs.
- Citations state which standard was violated, the gravity of the violation, the proposed penalty, and the abatement date (deadline to fix the hazard).
- Violation categories include: Willful (intentional violation or conscious indifference, carrying the highest penalties), Serious (high probability of death or serious physical harm), Other-Than-Serious, and Repeated.

> **★ HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
> *Memorize the inspection priorities. Imminent danger is always first. Understand that employers have 15 working days from receiving a citation to formally contest it in writing.*

Chapter 3: OSHA Recordkeeping and Reporting

29 CFR 1904

Employers with more than 10 employees in non-exempt industries must maintain rigorous records of work-related injuries and illnesses. This system tracks safety trends and guides OSHA's inspection targeting. Recordkeeping involves three primary forms: OSHA Form 300 (Log of Work-Related Injuries and Illnesses), OSHA Form 300A (Summary of Work-Related Injuries and Illnesses), and OSHA Form 301 (Injury and Illness Incident Report).

MANDATORY STANDARDS & TECHNICAL CORE

- An injury or illness is recordable if it results in death, days away from work, restricted work or transfer to another job, medical treatment beyond first aid, loss of consciousness, or a significant injury/illness diagnosed by a healthcare professional.
- First aid includes minor interventions like using bandages, ice packs, tweezers, or non-prescription medication at non-prescription strength; anything more advanced constitutes medical treatment.
- The OSHA 300A Annual Summary must be certified by a company executive and posted in a visible location from February 1st through April 30th of the following year.
- Severe incidents have explicit reporting timelines: Fatalities must be reported to OSHA within 8 hours. Any in-patient hospitalization, amputation, or loss of an eye must be reported within 24 hours.

> **★ HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
>
> *Distinguish between 'recording' a case on the log and 'reporting' to OSHA. Fatalities = 8 hours; Hospitalization/ Amputation/Eye Loss = 24 hours. The 300A must be posted Feb 1 - April 30.*

Chapter 4: Focus Four: Fall Hazards

29 CFR 1926 Subpart M

Falls represent the leading cause of fatalities in the construction industry, accounting for nearly one-third of all site deaths. OSHA's Focus Four initiative highlights these high-risk areas. In general construction, employers are required to provide fall protection whenever a worker is exposed to a fall of 6 feet or more to a lower level. Protection is also mandatory at any height when working above dangerous equipment or machinery.

MANDATORY STANDARDS & TECHNICAL CORE

- The three primary systems of fall protection are Guardrail Systems, Safety Net Systems, and Personal Fall Arrest Systems (PFAS).
- Guardrails must have a top rail height of 42 inches (plus or minus 3 inches), a midrail at roughly 21 inches, and a toeboard of at least 3.5 inches in height to prevent objects from falling. The top rail must withstand a downward/ outward force of 200 pounds.
- A Personal Fall Arrest System (PFAS) consists of an anchor point, an linking lanyard or deceleration device, and a full-body harness. Body belts are strictly prohibited for fall arrest.
- Anchorages used for PFAS must be independent of any platform support and capable of supporting at least 5,000 pounds per attached worker, or engineered to a safety factor of two.

> **★ HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
>
> *The magic trigger height for fall protection in construction is 6 feet. Know the guardrail dimensions (42" top, 21" mid) and the 5,000-lb anchor strength requirement.*

Chapter 5: Focus Four: Caught-In or -Between Hazards

29 CFR 1926 Focus Four

Caught-in or -between hazards are defined as incidents where a person is compressed, caught, crushed, or pinched between two or more objects. This focus area addresses severe structural and mechanical risks that result in asphyxiation, crushing, or traumatic amputations. Common scenarios include trench cave-ins, getting clothing or limbs entangled in unguarded machinery, and being crushed between a moving vehicle and a fixed structure.

MANDATORY STANDARDS & TECHNICAL CORE

- Trench cave-ins are the most prevalent caught-in/between fatal event on construction sites; a single cubic yard of soil can weigh as much as a car (approx. 3,000 lbs).
- Machine guarding is mandatory for all equipment with rotating, reciprocating, or moving components. Power transmission apparatus (belts, pulleys, shafts) must be fully enclosed.
- Lockout/Tagout (LOTO) protocols must be enforced during maintenance to prevent the accidental or unexpected energization of heavy equipment.
- Workers must maintain a safe distance from rotating heavy equipment, particularly the swing radius of excavators and cranes, which must be fully barricaded.

> **★ HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
> *Understand the difference between Struck-By and Caught-In. If a moving object hits a worker, it is Struck-By. If a worker is pinned between an object and a wall, or buried in a trench, it is Caught-In/Between.*

Chapter 6: Focus Four: Struck-By Hazards

29 CFR 1926 Focus Four

Struck-by hazards occur when a worker is injured by the impact of a flying, falling, swinging, or rolling object. Unlike caught-in situations where a worker is pinned, struck-by injuries result purely from the force of the kinetic impact. This represents a massive percentage of construction injuries, ranging from falling hand tools to heavy highway vehicles encroaching into work zones.

MANDATORY STANDARDS & TECHNICAL CORE

- Flying object hazards are common during grinding, chipping, sawing, and powder-actuated tool operations. Safety glasses and full-face shields are primary defenses.
- Falling objects are mitigated through the use of hard hats (ANSI Z89.1), toeboards on scaffolding, debris netting, and tool tethering protocols.
- Swinging objects typically involve rigging, crane loads, or materials suspended from excavators. Workers must never stand directly under suspended loads.
- Rolling objects include heavy equipment or vehicles moving on site. Heavy machinery must have functional backup alarms and horns, and spotters should be utilized in tight spaces.

> **★ HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
> *To prevent struck-by injuries from vehicles, workers must wear high-visibility vests (Class 2 or 3 depending on traffic speed) and heavy equipment must have working backup alarms or a dedicated spotter.*

Chapter 7: Focus Four: Electrocution Hazards

29 CFR 1926 Subpart K

Electrical current presents an invisible, immediate threat to construction workers. OSHA summarizes electrical safety with the acronym 'BE SAFE': Burns, Electrocution, Shock, Arc flash/blast, Fire, and Explosions. Shock occurs when a person becomes part of an electrical circuit, while electrocution is a fatal shock. Burns are the most common non-fatal injury resulting from electrical contact.

MANDATORY STANDARDS & TECHNICAL CORE

- Ground Fault Circuit Interrupters (GFCIs) are mandatory for all temporary 125-volt, single-phase, 15-, 20-, and 30-ampere receptacle outlets on construction sites. GFCIs detect current leakage (as low as 4-6 milliamperes) and interrupt power in milliseconds.
- Overhead power lines present an extreme electrocution risk. Equipment such as cranes, ladders, and scaffolding must maintain a absolute minimum clearance of 10 feet from power lines carrying up to 50kV.
- Extension cords must be rated for heavy-duty or extra-heavy duty use and must be inspected daily for cuts, exposed wires, or missing grounding prongs.
- Temporary lighting must be guarded to prevent accidental contact with bulbs, and cords must not be hung with staples or nails.

> **★ HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
> *GFCI protection is non-negotiable for temporary wiring. Remember the 10-foot clearance rule for power lines up to 50kV. GFCIs trigger at 4-6 mA.*

Chapter 8: Personal Protective Equipment (PPE)

29 CFR 1926 Subpart E

Personal Protective Equipment (PPE) is designed to protect workers from serious workplace injuries or illnesses resulting from contact with chemical, radiological, physical, electrical, mechanical, or other workplace hazards. Under the Hierarchy of Controls, PPE is explicitly classified as the last line of defense. Employers must first attempt to eliminate the hazard, substitute less dangerous materials, apply engineering controls, or enforce administrative rules.

MANDATORY STANDARDS & TECHNICAL CORE

- Employers are legally required to conduct a hazard assessment of the workplace to determine if PPE is necessary, and they must provide most PPE to employees at no cost.
- Head protection (hard hats) must meet ANSI Z89.1 standards and are classified as Type I (top impact) or Type II (top and lateral impact), with electrical classes G (General), E (Electrical, 20kV), and C (Conductive).
- Eye and face protection (safety glasses, face shields) must comply with ANSI Z87.1. Face shields alone do not provide adequate eye protection; they must be worn over safety glasses.
- Respiratory protection (29 CFR 1910.134) requires a written program, medical evaluations to ensure fitness, and annual fit testing before a worker can wear a respirator.

> **★ HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
> *PPE is the LAST line of defense. Hard hats must meet ANSI Z89.1. Employers must pay for standard PPE (except everyday items like standard cold-weather gear or ordinary steel-toe boots if transferable).*

Chapter 9: Health Hazards in Construction

29 CFR 1926 Subpart D

Health hazards in construction often cause delayed, chronic, and catastrophic illnesses rather than immediate physical injuries. These hazards include chemical exposures (dusts, fumes, vapors, gases), physical hazards (noise, vibration, radiation), biological hazards, and ergonomic stressors. Key chronic health threats on construction sites involve toxic mineral dusts and metals.

MANDATORY STANDARDS & TECHNICAL CORE

- Crystalline Silica exposure occurs during cutting, grinding, or drilling concrete, brick, and stone. The Permissible Exposure Limit (PEL) is 50 micrograms per cubic meter ($\mu g/m^3$) over an 8-hour TWA. Engineering controls like wet methods and HEPA vacuums are required.
- Asbestos is a fibrous mineral found in older insulation, tiling, and roofing. Disturbed asbestos releases microscopic fibers causing asbestosis, lung cancer, and mesothelioma. Strict containment and specialized training are required.
- Lead exposure is common during demolition or renovation of structures painted with lead-based paints. Lead disrupts blood-forming, nervous, and urinary systems, requiring blood lead level testing.
- Occupational noise exposure requires a continuous hearing conservation program when worker noise exposure equals or exceeds an 8-hour Time-Weighted Average (TWA) of 85 decibels (dBA).

> **★ HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
> *Silica PEL is 50 $\mu g/m^3$. Understand that health hazards focus on long-term (chronic) health damage, such as mesothelioma from asbestos or hearing loss from noise.*

Chapter 10: Hazard Communication (Globally Harmonized System - GHS)

29 CFR 1926.59 / 1910.1200

The Hazard Communication Standard (HazCom), often called the 'Right to Know' standard, was updated to align with the Global Harmonized System of Classification and Labelling of Chemicals (GHS), transitioning into a 'Right to Understand' standard. It ensures that the hazards of all chemicals produced or imported are evaluated and that information concerning their hazards is transmitted to employers and workers.

MANDATORY STANDARDS & TECHNICAL CORE

- Employers must maintain a comprehensive written Hazard Communication Program, an up-to-date chemical inventory, and ensure all containers are properly labeled.
- Safety Data Sheets (SDSs) must follow a strict, standardized 16-section format. They must be readily accessible to workers on all shifts.
- GHS shipping labels require six critical elements: Product Identifier, Supplier Identification, Signal Word ('Danger' for severe hazards, 'Warning' for less severe), Hazard Statements, Precautionary Statements, and Pictograms.
- There are 9 specific GHS pictograms enclosed in a red diamond border, indicating health hazards, flammability, toxicity, corrosiveness, or explosive properties.

> **★ HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
> *Know the number of sections on an SDS (16 sections). Understand the two primary GHS signal words: 'Danger' (high severity) and 'Warning' (lower severity).*

Chapter 11: Material Handling, Storage, Use, and Disposal

29 CFR 1926 Subpart H

Material handling encompasses moving, storing, and disposing of construction materials. Improper handling leads to structural collapses, struck-by incidents, and severe musculoskeletal injuries. This standard establishes strict rules for stacking heights, structural loading capacities, and rigging safety for material handling equipment.

MANDATORY STANDARDS & TECHNICAL CORE

- Materials stored inside buildings under construction must not be placed within 6 feet of any hoistway or inside floor openings, nor within 10 feet of an exterior wall that does not extend above the top of the material.

- Lumber piles must not exceed 16 feet in height for manual handling, or 20 feet if handled with a forklift. Bricks must not be stacked higher than 7 feet, and must be tapered after 4 feet.
- Rigging equipment (slings, shackles, hooks, chains) must be inspected by a competent person prior to every shift. Damaged or defective rigging must be immediately cut and discarded.
- An enclosed chute must be used whenever materials are dropped more than 20 feet to any point outside the exterior walls of a building.

> **★ HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
> *Remember the 20-foot rule for disposal chutes. Know stacking maximums (Lumber manual: 16ft, Forklift: 20ft; Brick: 7ft max).*

Chapter 12: Hand and Power Tools Safety

29 CFR 1926 Subpart I

Hand and power tools are a source of constant danger on site due to cutting edges, high velocity components, and extreme pressure. Hazards include flying projectiles, electrical shocks, thermal burns, and puncture wounds. The standard mandates proper maintenance, guarding of all moving parts, and specific power controls based on tool design.

MANDATORY STANDARDS & TECHNICAL CORE

- All hand-held power tools must be equipped with appropriate safety switches. Hand-held circular saws, chainsaws, and percussion tools require a constant-pressure 'dead-man' switch that shuts off power when released.
- Belts, gears, shafts, pulleys, sprockets, and spindles of power transmission apparatus must be fully guarded if they are within 7 feet of the working floor or platform.
- Pneumatic tools must be secured to the hose by a positive locking device or safety clip to prevent the tool attachment from being accidentally ejected.
- Powder-actuated tools operate like a loaded firearm. Operators must be trained and licensed. Tools must never be loaded until immediately before use and never pointed at any worker.

> **★ HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
> *Know that a 'dead-man' switch is a constant-pressure switch. Understand the safety clips required on pneumatic hoses, and that power transmission parts below 7 feet must be guarded.*

Chapter 13: Welding, Cutting, and Brazing Safety

29 CFR 1926 Subpart J

Welding, cutting, and brazing (collectively called Hot Work) introduce high-voltage electrical currents, intense radiant energy (UV/IR radiation), toxic fumes, and open flames. The risk of fire and acute or chronic health damage is severe. Proper storage of compressed gas cylinders and fire prevention are the primary focus areas of this standard.

MANDATORY STANDARDS & TECHNICAL CORE

- Compressed gas cylinders must be stored upright, secured to prevent tipping, and have their valve protection caps tightly in place when not in active use.
- Oxygen cylinders in storage must be separated from fuel-gas cylinders (such as acetylene) or combustible materials by a minimum distance of 20 feet or by a noncombustible barrier at least 5 feet high with a fire-resistance rating of at least 30 minutes.

- A dedicated 'Fire Watch' must be maintained for at least 30 minutes after completion of welding or cutting operations to detect and extinguish any smoldering or hidden fires.
- Workers must wear specialized eye protection with specific filter lens shade numbers (e.g., Shade 10-14 for arc welding) to prevent 'arc eye' or permanent retinal burns.

> **★ HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
> *Memorize the cylinder storage separation rule: 20 feet apart OR a 5-foot high wall with a 30-minute fire rating. Fire watch must stay for at least 30 minutes post-work.*

Chapter 14: Scaffolding Safety

29 CFR 1926 Subpart L

Scaffolding provides elevated work platforms but introduces catastrophic fall and structural collapse hazards. OSHA requires that scaffolds be designed by a qualified person and erected, moved, dismantled, or altered under the direct supervision and guidance of a designated Competent Person. This chapter covers the mechanical, structural, and operational limits of temporary platforms.

MANDATORY STANDARDS & TECHNICAL CORE

- Each scaffold and its components must be capable of supporting, without failure, its own weight and at least 4 times the maximum intended load (4x safety factor).
- Fall protection (guardrails or personal fall arrest systems) is mandatory for any employee on a scaffold platform that is more than 10 feet above a lower level.
- Scaffold platforms must be fully planked or decked between the front uprights and the guardrail supports. Planks must overlap by at least 12 inches unless cleated, and extend past their supports by 6 to 12 inches.
- The clearance between a scaffold and overhead power lines must be maintained at a minimum of 10 feet for lines up to 50kV to eliminate electrocution vectors.

> **★ HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
> *Fall protection on scaffolds is triggered at 10 feet (unlike the general 6 feet for construction). Scaffold capacity must have a 4x safety factor.*

Chapter 15: Cranes, Derricks, Hoists, Elevators, and Conveyors

29 CFR 1926 Subpart CC

Heavy lifting equipment poses extreme structural and striking hazards. Cranes must be engineered, positioned, operated, and maintained with absolute precision. Incidents involve tip-overs due to unstable ground, mechanical failure, or striking workers with moving booms or suspended loads. This standard details structural certifications, inspection timelines, and personnel qualifications.

MANDATORY STANDARDS & TECHNICAL CORE

- Cranes must be set up on firm, level ground. Outriggers must be fully extended and set on stable cribbing or pads. The crane must be completely leveled before any lift.
- The standard mandates certified crane operators, qualified riggers (to secure loads), and qualified signal persons (to direct the operator using standard hand or radio signals).
- A competent person must conduct visual inspections of the crane before every single shift. Thorough, written annual inspections must be performed and recorded.
- Load charts detailing the exact lifting capacity of the crane at various boom angles and radii must be clearly visible and accessible inside the crane cab.

> **★ HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
> *Remember that operators must be certified, and load charts must always be inside the cab. Power line clearance remains 10 feet for lines up to 50kV unless line is de-energized.*

Chapter 16: Excavations and Trenching Safety

29 CFR 1926 Subpart P

Excavation and trenching work is among the most hazardous operations in construction. An excavation is any man-made cut, cavity, trench, or depression formed by earth removal. A trench is a narrow excavation where the depth is greater than the width, and the width does not exceed 15 feet. Cave-ins present an immediate, fatal threat of suffocation and crushing.

MANDATORY STANDARDS & TECHNICAL CORE

- Daily inspections of excavations, adjacent areas, and protective systems must be conducted by a designated Competent Person before work begins and after any rain or hazard-increasing event.
- Protective systems against cave-ins are mandatory for all excavations 5 feet deep or greater, or any depth if the competent person identifies unstable soil. Systems include Sloping/Benching, Shoring (hydraulic or timber), and Shielding (trench boxes).
- A safe means of egress (ladders, steps, or ramps) is required in trenches 4 feet deep or more, and must be located so that a worker does not have to travel more than 25 feet laterally to reach it.
- Excavated materials (spoils) and heavy equipment must be kept at a minimum distance of 2 feet back from the edge of the excavation to prevent surcharge loads from triggering cave-ins.

> **★ HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
> *This is a heavy exam topic. Protective systems required at 5 feet deep. Ladders/egress required at 4 feet deep, and within 25 feet of lateral travel. Spoils must be 2 feet back.*

Chapter 17: Concrete and Masonry Construction

29 CFR 1926 Subpart Q

Concrete and masonry operations involve massive dead-weight loads, temporary structural framing, and specialized tool usage. Hazards include premature formwork removal, structural collapse of unbraced masonry walls, and skin contact with wet concrete (caustic chemical burns). This standard ensures structural integrity during the critical curing phase.

MANDATORY STANDARDS & TECHNICAL CORE

- Formwork must be engineered, fabricated, and braced to safely withstand all vertical and lateral loads. Shoring must be inspected immediately prior to, during, and after concrete placement.
- No employee is permitted to stand or work under concrete buckets while they are being elevated or routed into position.
- All protruding reinforcing steel (rebar) onto or into which employees could fall must be guarded to eliminate the hazard of impalement. High-impact square rebar caps or wood troughs are standard.
- A Limited Access Zone must be established whenever a masonry wall is under construction. The zone must equal the height of the wall plus 4 feet, and run the entire length of the wall on the unbraced side.

> ★ **HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
> *Protruding rebar must be guarded if it poses an impalement hazard. Understand the Limited Access Zone dimensions for masonry: Height of wall + 4 feet.*

Chapter 18: Stairways and Ladders

29 CFR 1926 Subpart X

Stairways and ladders provide vital vertical access on a construction site but are frequent sources of falls and serious injuries due to improper use, structural degradation, or carrying heavy loads. This standard details precise mechanical and dimensional parameters for ladders, step heights, and handrail systems.

MANDATORY STANDARDS & TECHNICAL CORE

- A stairway or ladder must be provided at all personnel access points where there is a break in elevation of 19 inches or more and no ramp, runway, or sloped embankment is available.
- Extension ladders must extend at least 3 feet (36 inches) above the upper landing surface to ensure a secure handhold, or the ladder must be secured at its top with a grab rail provided.
- Non-self-supporting ladders (extension/straight ladders) must be positioned at a 4:1 ratio: for every 4 feet of vertical height, the base of the ladder must be placed 1 foot out from the structure.
- Stairways having four or more risers or rising more than 30 inches, whichever is less, must be equipped with at least one handrail and a stair rail system along each unprotected edge.

> ★ **HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
> *Know the 19-inch elevation rule for requiring a ladder/stairway. Remember the 4:1 ladder ratio and the 3-foot extension requirement above the landing platform.*

Chapter 19: Confined Space Entry in Construction

29 CFR 1926 Subpart AA

Confined spaces present hidden, lethal environments. A confined space is defined as an area that is large enough for an employee to enter and perform work, has limited or restricted means for entry or exit, and is not designed for continuous employee occupancy. A Permit-Required Confined Space (PRCS) contains additional, recognizable operational hazards.

MANDATORY STANDARDS & TECHNICAL CORE

- A Permit-Required Confined Space has one or more of the following: Contains or has potential to contain a hazardous atmosphere; contains material with potential to engulf an entrant; has internal configuration that could trap an entrant; or contains any other recognized serious safety hazard.
- Atmospheric testing is mandatory before entry and must test for hazards in a specific order: 1st Oxygen content (safe range 19.5% to 23.5%), 2nd Flammable gases/vapors, and 3rd Toxic air contaminants.
- Entry requires a dedicated, trained team: the Authorized Entrant (enters the space), the Attendant (remains outside, monitors, and never enters unless relieved), and the Entry Supervisor (authorizes and signs the permit).
- Continuous mechanical ventilation and continuous atmospheric tracking are primary engineering controls used to maintain a safe environment.

> ★ **HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
> *Know the entry team roles (Entrant, Attendant, Supervisor). Remember the order of atmospheric testing: Oxygen first, then Flammability, then Toxicity. Safe oxygen levels are 19.5% - 23.5%.*

Chapter 20: Fire Protection and Prevention

29 CFR 1926 Subpart F

Construction sites contain concentrated configurations of fuel sources, ignition vectors, and variable escape routing. Employers must maintain a comprehensive fire protection and prevention program throughout all phases of construction, ensuring firefighting equipment is accessible and emergency action plans are understood.

MANDATORY STANDARDS & TECHNICAL CORE

- A fire extinguisher rated not less than 2A must be provided for every 3,000 square feet of the protected building area. The maximum travel distance from any point of the protected area to the nearest fire extinguisher must not exceed 100 feet.
- In multi-story buildings, at least one fire extinguisher must be located on each floor, positioned adjacent to the primary stairway.
- Flammable and combustible liquids must be stored only in approved, closed containers (such as safety cans with self-closing lids and flash-arresting screens).
- No more than 25 gallons of flammable or combustible liquids may be stored outside of an approved, designated fire storage cabinet.

> **★ HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
> *Remember the 3,000 sq ft rule and the 100-foot maximum travel distance to a fire extinguisher. One extinguisher is mandatory on every floor of a multi-story project.*

Chapter 21: Ergonomics in Construction

General Duty / Best Practices

Ergonomics focuses on fitting the job to the worker rather than forcing the worker's body to adapt to the job. Construction tasks frequently involve heavy manual lifting, awkward body postures, high repetition, and severe vibration from tools. Over time, these physical stressors lead to Musculoskeletal Disorders (MSDs) such as back strains, tendonitis, and carpal tunnel syndrome.

MANDATORY STANDARDS & TECHNICAL CORE

- Musculoskeletal Disorders account for a substantial percentage of lost-time injuries in construction, affecting muscles, nerves, tendons, ligaments, and spinal discs.
- Ergonomic controls target high-risk factors: implementing mechanical lifting devices (forklifts, material hoists, vacuum lifters) to eliminate manual strain.
- When manual lifting is unavoidable, workers must use safe lifting techniques: keeping the load close to the body, lifting with the legs instead of the back, and avoiding twisting movements.
- Administrative controls include job rotation, scheduled stretch-and-flex breaks, and assigning team lifts for heavy or awkwardly shaped materials exceeding 50 pounds.

> **★ HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
> *Ergonomics relies heavily on the hierarchy of controls. Mechanical assists are engineering controls, while team lifting and job rotation represent administrative controls.*

Chapter 22: Motor Vehicles, Mechanized Equipment, and Marine Operations

29 CFR 1926 Subpart O

The interaction of heavy mobile machinery and pedestrian workers creates an operational environment with zero margin for error. Subpart O covers earthmoving equipment, dump trucks, agricultural tractors, and commercial vehicles operating within site boundaries. This standard establishes operating parameters to prevent tip-overs and run-overs.

MANDATORY STANDARDS & TECHNICAL CORE

- All vehicles operating on a construction site must have a functional service brake system, emergency brakes, audible horns, and operational brake lights.
- Heavy earthmoving equipment (bulldozers, loaders, scrapers) must be equipped with Rollover Protective Structures (ROPS) and seatbelts. Operators must wear seatbelts at all times during operation.
- No employer may use any motor vehicle or earthmoving equipment that has an obstructed view to the rear unless the vehicle has a reverse signal alarm audible above surrounding noise, or a spotter signals that it is safe to back up.
- When heavy equipment is left unattended, the operator must fully lower all blades, buckets, or bowls to the ground, apply the parking brake, and turn off the engine.

> **★ HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
> *Vehicles with an obstructed rear view MUST have a functional backup alarm or utilize a dedicated spotter. Seatbelts are mandatory on ROPS-equipped machinery.*

Chapter 23: Demolition Safety

29 CFR 1926 Subpart T

Demolition work involves dismantling or wrecking a structure, introducing unpredictable structural behaviors and hidden material hazards. Because components are being weakened intentionally, structural stability is highly volatile. Subpart T enforces planning and preparatory procedures to protect workers from unplanned collapses.

MANDATORY STANDARDS & TECHNICAL CORE

- Prior to starting any demolition operations, a qualified person must conduct a comprehensive engineering survey of the structure to determine its condition and the possibility of an unplanned collapse.
- All electric, gas, water, steam, sewer, and chemical chemical utility lines must be shut off, capped, or otherwise controlled outside the structure line before work begins.
- Adjoining structures must be evaluated and shored or braced to guarantee they are not structurally compromised by the demolition work.
- Chutes or floor holes used to drop debris must be completely barricaded with guardrails, and lower discharge areas must be completely closed off to prevent struck-by injuries.

> **★ HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
> *The engineering survey is the absolute first step in demolition safety and must be conducted by a qualified person and documented.*

Chapter 24: Blasting and the Use of Explosives

29 CFR 1926 Subpart U

The storage, transport, handling, and detonation of explosive materials on a construction site are subject to severe regulations. Blasting operations introduce massive concussion, flying debris, and accidental detonation risks. Subpart U ensures that only highly trained, licensed professionals control explosive charges.

MANDATORY STANDARDS & TECHNICAL CORE

- Only authorized and properly licensed individuals (Blasters) are permitted to handle, transport, or detonate explosive materials on a site.
- Explosives must be stored in specialized 'Magazines' that are locked, bullet-resistant, fire-resistant, weather-resistant, and ventilated, placed away from active work tracks.
- A strict audible signaling system must be established and communicated to all personnel prior to a blast: Warning Signal (1-minute series of long blasts), Blast Signal (series of short blasts), and All-Clear Signal (prolonged single blast).
- Blasting operations must cease immediately upon the approach or detection of an electrical storm, lightning, or stray radio frequencies that could trigger electronic blasting caps.

> **★ HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
> *Understand the blasting signaling sequence (Warning = long blasts; Blast = short blasts; All-Clear = single long blast). Operations must halt for lightning or stray radio signals.*

Chapter 25: Electrical Safety (General & Distribution Systems)

29 CFR 1926 Subpart K (Wiring & Systems)

Beyond basic user tool safety, the structural distribution of power across a construction site involves complex infrastructure. Temporary wiring, main distribution boards, panelboards, and branch circuits must be designed to withstand rugged physical environmental conditions while preventing arc flash and ground faults.

MANDATORY STANDARDS & TECHNICAL CORE

- Temporary electrical branch circuits must originate from an approved, enclosed power distribution panelboard. Wiring must be protected from physical damage, sharp edges, and vehicular traffic.
- As an alternative to GFCIs, employers may implement an Assured Equipment Grounding Conductor Program (AEGCP). This requires a detailed written description of the site-specific testing procedures.
- The AEGCP requires continuous visual inspection of all cords and equipment daily, plus formal continuity and grounding tests conducted and logged every 3 months by a competent person.
- All lamps for temporary lighting must be enclosed in protective cages to prevent accidental breaking or skin contact with hot surfaces; temporary lights must never be suspended by their power cords.

> **★ HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
> *The AEGCP requires a written program and formal testing every 3 months. Know that temporary lights must be guarded with cages and cannot be hung by their power cords.*

Chapter 26: Process Safety Management & HAZWOPER Overview

29 CFR 1910.119 / 1926.65

Construction activities frequently take place inside or adjacent to active chemical plants, refineries, or hazardous waste sites. Process Safety Management (PSM) prevents or minimizes the consequences of catastrophic releases of toxic, reactive, flammable, or explosive chemicals. HAZWOPER governs construction operations involving hazardous substance cleanups or emergency response.

MANDATORY STANDARDS & TECHNICAL CORE

- PSM standard applies to processes containing highly hazardous chemicals above specific threshold quantities. Contractors working on these sites must be trained in the specific safety and emergency procedures of that facility.

- Host employers and contractors must formally share information regarding site hazards, hot work permits, and emergency evacuation protocols.
- HAZWOPER (Hazardous Waste Operations and Emergency Response) requires specialized training (24-hour or 40-hour levels) for construction workers performing cleanup operations.
- HAZWOPER operations require strict site control zones: the Hot Zone (Exclusion Zone/Contaminated Area), the Warm Zone (Contamination Reduction Zone), and the Cold Zone (Support Zone).

> **★ HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
> *Understand the three HAZWOPER site control zones: Hot (contaminated area), Warm (decontamination area), and Cold (clean support area).*

Chapter 27: Safety and Health Programs & Safety Culture

OSHA Recommended Practices

Modern safety science emphasizes that reactive compliance is insufficient to prevent workplace incidents. OSHA's Recommended Practices for Safety and Health Programs guide employers in transitioning to proactive, systematic hazard management. Building a robust safety culture reduces incident rates, improves worker morale, and ensures operational compliance.

MANDATORY STANDARDS & TECHNICAL CORE

- A successful safety program relies on seven core elements: Management Leadership, Worker Participation, Hazard Identification, Hazard Prevention/Control, Education/Training, Evaluation/Improvement, and Multi-Employer Site Coordination.
- Management leadership involves committing financial and structural resources, establishing clear safety policies, and leading by example.
- Worker participation ensures that frontline employees are actively involved in safety committees, hazard identification, and workplace inspections without fear of retaliation.
- Programs must distinguish between Laging Indicators (reactive metrics such as Incident Rates, injuries) and Leading Indicators (proactive metrics such as safety training hours, hazard corrections, and audits).

> **★ HIGH-YIELD EXAM FOCUS & DIAGNOSTICS:**
> *Differentiate leading and lagging indicators. Leading indicators are proactive (e.g., training, audits); lagging indicators are reactive (e.g., injury rates, workers' comp costs).*

💡 High-Probability Exam Pass Strategies

- **Watch the Trigger Numbers:** OSHA exam questions love specific numeric thresholds. Memorize: 6 feet (General Construction Fall Protection), 10 feet (Scaffold Fall Protection & Power Line Clearance), 4 feet (Trench Ladder requirement), 5 feet (Trench Protective System requirement), and 2 feet (Trench Spoils setback).
- **Hierarchy of Controls:** If a question asks for the 'best' or 'first' safety method, always look for Elimination or Engineering Controls (like guardrails or ventilation) before looking for PPE (like harness or mask). PPE is always the last resort.
- **Reporting Window Triggers:** Understand the absolute difference between 8 hours (Fatalities) and 24 hours (Hospitalization, Amputation, or Eye Loss). Missing these triggers is a standard trap on safety exams.
- **Competent vs. Qualified:** A 'Competent Person' has the authority to stop work and identify hazards. A 'Qualified Person' has a technical degree or certificate (like a structural engineer designing a scaffold or crane lift plan).